Mathematical Modeling Lecture Notes
Markov Chains

I. Introductory example: The leading brewery on the West Coast, A, is concerned about its market position relative to its major competitor, B, and to its other competitors, C. Their analysts believe that, in the short run, the major changes to the number of customers are due to brand switching. In particular, research indicates that, from month-to-month, 70% of A's customers stick with A, 25% switch to B, and the remainder switch to C. Of B's customers, 20% switch to A, 75% remain with B, and 5% switch to C. Of the customers in category C, 10% switch to A, 10% switch to B, and 80% remain with C. What happens to the evolution of this system over time?
   a. Since there are a finite number of individuals that neither enter nor leave the system, we can model this situation with a state transition diagram.

   b. We have the three coupled difference equations:
      i. $a_{n+1} = 0.7a_n + 0.2b_n + 0.1c_n$.
      ii. $b_{n+1} = 0.25a_n + 0.75b_n + 0.1c_n$.
      iii. $c_{n+1} = 0.05a_n + 0.05b_n + 0.8c_n$.
   c. Show evolution over time using Python.
   d. It appears that the final population ratios are independent of the initial population.
   e. Matrix form
      i. We can interpret these fractions as the probabilities that a particular customer will switch from one brand to another.
      ii. So if p_{ij} is the probability that an individual switches from brand i to brand j, then, in matrix form, this becomes $P = \begin{bmatrix} 0.7 & 0.25 & 0.05 \\ 0.2 & 0.75 & 0.05 \\ 0.1 & 0.1 & 0.8 \end{bmatrix}$.
      iii. Note that the probability transition matrix is the transpose of the matrix we use in Python to represent the evolution of the system state over time.
      iv. We will be focusing on P rather than the equation form we saw before.
   f. This is an example of a Markov chain.

II. Basics and terminology
   a. Stochastic processes
      i. A *stochastic process*, $\{X_t\}_{t \in T}$, is an indexed sequence of random variables.
      ii. Often the random variables are indexed by time, so that you have a sequence of random variables representing the state of some system over time.
      iii. Examples
         1. Stock prices over time
         2. Number of available parking spaces in the Thompson parking lot every hour

3. State of a machine (working, broken-down, or being repaired) in a manufacturing plant

b. A *Markov chain* is a stochastic process with the property that $P(X_{t+1} = j|X_0 = k_0, X_1 = k_1, \ldots, X_{t-1} = k_{t-1}, X_t = i) = P(X_{t+1} = j|X_t = i)$.
   i. In other words, the conditional distribution of a future event given the past states of the system and the present state of the system is dependent only on the present.
   ii. Alternatively, given the current state of the system the past is irrelevant.
   iii. Examples
      1. Brewery switching, at least the way we modeled it, would be a Markov chain.
      2. Stock prices over time
         - Probably not
         - However, if you define the "present" state so that it goes far enough back in time you might be able to model this as a Markov chain.
      3. Available parking spaces: probably depends on time of day as well as how many spaces available in the previous time period. (So this would be a non-stationary Markov chain.)
      4. Machine state: This might reasonably be modeled as a Markov chain.
      5. The children's game "Chutes and Ladders"

c. Transition probabilities
   i. Let $S = \{0, 1, \ldots, M\}$ denote the set of possible system states.
   ii. Then $P(X_{t+1} = j|X_t = i)$, $i, j \in S$, is called a *transition probability.*
   iii. If, for each i and j, $P(X_{t+1} = j|X_t = i) = P(X_1 = j|X_0 = i)$ for all t then the transition probabilities are *stationary.*
      1. In other words, the stationarity property means that the probabilities are not time-dependent.
      2. We will be focusing on problems in which the transition probabilities are stationary.
   iv. The n-step transition probabilities
      1. Stationarity implies, for each i, j, and n, $P(X_{t+n} = j|X_t = i) = P(X_n = j|X_0 = i)$ for all t.
      2. These are called the *n-step transition probabilities.*
   v. Simplifying the notation
      1. We denote $P(X_{t+1} = j|X_t = i)$ by p_{ij}.
      2. We denote $P(X_{t+n} = j|X_t = i)$ by $p_{ij}^{(n)}$.
      3. We don't need to worry about t because of the stationarity.
      4. We often represent the n-step transition probabilities in matrix form:
      $$\mathbf{P}^{(n)} = \begin{bmatrix} p_{00}^{(n)} & p_{01}^{(n)} & \cdots & p_{0M}^{(n)} \\ p_{10}^{(n)} & p_{11}^{(n)} & \cdots & p_{1M}^{(n)} \\ \vdots & \vdots & \ddots & \vdots \\ p_{M0}^{(n)} & p_{M1}^{(n)} & \cdots & p_{MM}^{(n)} \end{bmatrix}.$$
   vi. Some properties
      1. Non-negativity: $p_{ij}^{(n)} \geq 0$ for all i, j
      2. $\sum_{j=0}^{M} p_{ij}^{(n)} = 1$ for each i

d. Gambling example: Consider a gambler who, at each play of a game, either wins \$1 with probability p or loses \$1 with probability $1-p$. The gambler continues playing until he either goes broke or attains a fortune of \$$N$.
   i. This is a Markov chain.
   ii. One-step transition probabilities
      1. $p_{i,i+1} = p$, $p_{i,i-1} = 1-p$ for $1 \leq i \leq N-1$
      2. $p_{00} = p_{NN} = 1$
      3. All other one-step probabilities are 0.
   iii. In matrix form this is $\begin{bmatrix} 1 & 0 & 0 & 0 & \cdots & 0 \\ 1-p & 0 & p & 0 & \cdots & 0 \\ 0 & 1-p & 0 & p & \cdots & 0 \\ 0 & 0 & 1-p & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & 1 \end{bmatrix}$.

III. Calculating the transition probabilities
   a. How do you calculate the n-step transition probabilities from the one-step transition probabilities?
      • These are useful because they allow you to see how the system behaves in the long run.
   b. Matrix representation
      i. If we take the case $n = 2$ we obtain $p_{ij}^{(2)} = \sum_{k=0}^{M} p_{ik}p_{kj}$.
         1. But this is just the usual matrix multiplication operation.
         2. Thus $\mathbf{P}^{(2)} = \mathbf{P}^2$.
      ii. Similarly, $p_{ij}^{(3)} = \sum_{k=0}^{M} p_{ik}^{(2)}p_{kj} = \sum_{k=0}^{M} p_{ik}p_{kj}^{(2)}$.
         • This implies $\mathbf{P}^{(3)} = \mathbf{P}^2\mathbf{P} = \mathbf{P}\mathbf{P}^2 = \mathbf{P}^3$.
      iii. A straightforward induction proof shows that $\mathbf{P}^{(n)} = \mathbf{P}^n$.
   c. Back to the brewery example
      i. This means that we can calculate the probability that after n months someone will have switched to any other brand simply by taking powers of the one-step transition matrix.
      ii. It doesn't take too long, by continually taking powers of the matrix, to obtain $\begin{bmatrix} 0.36 & 0.44 & 0.2 \\ 0.36 & 0.44 & 0.2 \\ 0.36 & 0.44 & 0.2 \end{bmatrix}$ repeating itself.
      iii. This means that, for instance, the probability of starting with brand i in time 0 and ending with brand A in the long run, regardless of the value of i, is 0.36.
      iv. Thus, in the long run, 36% of the people drink brand A, 44% of the people drink brand B, and 20% of the people drink brand C.

IV. Classification of states
   a. State j is said to be *accessible* from state i if $p_{ij}^{(n)} > 0$ for some $n \geq 0$.
      i. In other words, j is accessible from i if it is at all possible to be in state j after once being in state i.
      ii. Examples
         1. In the brewery example, all states are accessible from all other states.
         2. In the gambling example, state 1 is not accessible from state 0.
   b. Two states i and j are said to *communicate* if they are accessible from each other.
      i. Some properties of the communication relation

1. Since $p_{ii}^{(0)} = 1$, every state communicates with itself. (reflexive property)
2. If state i communicates with state j then state j communicates with state i. (symmetric property)
3. If state i communicates with state j and state j communicates with state k then state i communicates with state k. (transitive property)
4. This means that "communication" is an equivalence relation.

ii. Since communication is an equivalence relation the set of states can be divided into distinct classes with the property that every state is accessible from every other state in its class.

iii. If there is only one class the Markov chain is said to be *irreducible*.

iv. Examples
1. The Markov chain in the brewery example is irreducible.
2. The Markov chain in the gambling example has three classes of states: $\{0\}$, $\{N\}$, and $\{1, 2, \ldots, N-1\}$.

c. A state is said to be *transient* if, once the state has been reached, there is a positive probability that the state will not be reached again.

i. Alternatively, the state i is transient if there exists some state j such that j is accessible from state i but state i is not accessible from state j.

ii. A state that is not transient is *recurrent*; i.e., once the state has been reached there is probability 1 that it will be reached again.

iii. Examples
1. Every state in the brewery example is recurrent.
2. The two states 0 and N in the gambling example are recurrent; all other states are transient.

iv. Long-term visits to recurrent and transient states
1. How many times is a recurrent state visited?
- Suppose the process enters the recurrent state i.
- Since i is recurrent it will be visited again with probability 1.
- Once it's visited again the fact that it's recurrent means that it will be visited a third time.
- Thus if a recurrent state is visited once it is visited infinitely often.

2. What's the probability that, if a transient state is visited, it is visited at least n times?
- Suppose the process enters the transient state i.
- There is a positive probability q that i will not be visited again.
- So the process either visits i a second time with probability $1 - q$, or it doesn't visit i again with probability q.
- Thus the probability that the process visits i a third time is $(1 - q)^2$.
- The probability that the process visits i an nth time is $(1 - q)^{n-1}$.
- So the probability that the process visits i infinitely many times is 0.
- With probability 1, then, the process visits i only finitely many times.
- In fact, if the process once visits state i the probability that it visits i exactly n times is $q(1 - q)^{n-1}$.
- Thus the number of times that, once visited, the process visits i is $geometric(q)$.

v. Some properties
1. Questions

- Can all states be recurrent?
- Can all states be transient?
- Can you have transient and recurrent states in the same communication class?

2. As the process is infinite, not all states in a finite-state Markov chain can be transient.
3. Since not all the states in a finite-state Markov chain can be transient, at least one must be recurrent.
4. If i and j are in the same class and i is recurrent then j must be as well.
   - Every time the process reaches i there is a positive probability p that the process will reach j before reaching i again. (This is because j is accessible from i.)
   - The probability that the process reaches i n more times before reaching j again is thus $(1-p)^n$.
   - Taking the limit, the probability that the process does not reach j again after the first visit to i is 0.
   - By the Markov property, this argument holds for all visits to i.
   - Thus the process must reach j infinitely many times.
   - Therefore, j is recurrent.
5. Similarly, if i and j are in the same class and i is transient then j must be as well. (Ask why this makes sense.)
6. Recurrence and transience are class properties.
7. Thus all states in an irreducible, finite-state Markov chain are recurrent.
8. An infinite-state Markov chain can have all states transient. The random walk on $\mathbb{R}^3$ is an example.

vi. A state i is said to be *absorbing* if $p_{ii} = 1$.
1. Thus a state is absorbing if the process never leaves it once it enters it.
2. Examples
   - The Markov chain in the brewery example has no absorbing states.
   - States 0 and N in the gambling example are both absorbing states.

vii. Example: Determine the transient and recurrent states in the following Markov chain.

$$\mathbf{P} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 \\ 0 & 0 & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 0 & \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{4} & \frac{1}{4} & 0 & 0 & \frac{1}{2} \end{bmatrix}$$

1. States 0 and 1 communicate with each other, as do states 2 and 3.
2. However, no states communicate with state 4, even though states 1 and 2 are accessible from state 4.
3. We have the recurrent classes $\{0, 1\}$ and $\{2, 3\}$ and the transient class $\{4\}$.

d. Periodicity

i. The *period* of the state i is the value of t, $t > 1$, such that $p_{ii}^{(n)} = 0$ for all values of n other than $t, 2t, 3t, \ldots$, and t is the largest such integer with this property.

ii. Text definition: "If there are two consecutive numbers s and $s+1$ such that the process can be in state i at times s and $s+1$, the state is said to have period 1 and is called an *aperiodic* state."

iii. Examples
1. The three states in the brewery example are aperiodic.
2. States $1, 2, \ldots, N-1$ in the gambling example all have period 2.
3. States 0 and N in the gambling example are aperiodic.

iv. Periodicity turns out to be a class property.

v. Ergodic
1. Aperiodic states in a finite-state Markov chain are said to be *ergodic*.
2. A Markov chain is said to be *ergodic* if all of its states are ergodic.
3. Irreducible, ergodic Markov chains form an important class of Markov chains (partly because of their nice properties).

V. Long-run properties

a. Steady-state probabilities

i. Theorem: For an irreducible ergodic Markov chain $\lim_{n\to\infty} p_{ij}^{(n)}$ exists and is independent of i. Moreover, we have $\lim_{n\to\infty} p_{ij}^{(n)} = \pi_j > 0$, and if $\pi = \begin{bmatrix}\pi_0 & \pi_1 & \cdots & \pi_M\end{bmatrix}$ then π is the unique solution to $\pi\mathbf{P} = \pi$, $\sum_{j=0}^{M} \pi_j = 1$.
1. The proof of this theorem is beyond the scope of this course.
2. The result says both that 1 is a left eigenvalue and that π is a corresponding left eigenvector of the matrix $\mathbf{P}$.
3. Alternatively, 1 and π^T are an eigenvalue and a corresponding eigenvector, respectively, of $\mathbf{P}^T$.

ii. The π_j's are called the *steady-state probabilities* of the Markov chain, as π_j is the probability that, in the long run, the stochastic process will be in state j.
1. This explains the $\pi\mathbf{P} = \pi$ equation.
2. In general, if the row vector $\mathbf{v}_t$ contains the probabilities of being in the various states at time t, then, because $\mathbf{P}$ is the one-step transition matrix, the row vector $\mathbf{v}_{t+1} = \mathbf{v}_t\mathbf{P}$ contains the probabilities of being in the various states at time $t+1$.
3. With this in mind, if $\mathbf{v}_t$ is equal to the steady-state probability vector π, then $\mathbf{v}_{t+1}$ must be the same as $\mathbf{v}_t$ (i.e., the probabilities of being in the various states can't be any different in the next time period).
4. Hence $\pi\mathbf{P} = \pi$.

iii. Example: Find the steady-state vector for the brewery Markov chain.
1. The Markov chain here is irreducible and ergodic; thus the theorem applies.
2. We have $\mathbf{P} = \begin{bmatrix}0.7 & 0.25 & 0.05\\ 0.2 & 0.75 & 0.05\\ 0.1 & 0.1 & 0.8\end{bmatrix}$.
3. So we need to solve the following system of equations:
   - $\pi_A = 0.7\pi_A + 0.2\pi_B + 0.1\pi_C$
   - $\pi_B = 0.25\pi_A + 0.75\pi_B + 0.1\pi_C$
   - $\pi_C = 0.05\pi_A + 0.05\pi_B + 0.8\pi_C$
4. This becomes:
   - $0 = -0.3\pi_A + 0.2\pi_B + 0.1\pi_C$
   - $0 = 0.25\pi_A - 0.25\pi_B + 0.1\pi_C$
   - $0 = 0.05\pi_A + 0.05\pi_B - 0.2\pi_C$
5. The solution to this system turns out to be underdetermined, with $\pi_A = 9/5\pi_C$ and $\pi_B = 11/5\pi_C$.

6. However, we must have $\pi_A + \pi_B + \pi_C = 1$.
7. Thus we can take the vector $\begin{bmatrix} 9 & 11 & 5 \end{bmatrix}$ and scale it.
8. This yields $\pi = \begin{bmatrix} 9/25 & 11/25 & 5/25 \end{bmatrix} = \begin{bmatrix} 0.36 & 0.44 & 0.2 \end{bmatrix}$.
9. Alternatively, we could have asked Python (or some other technology program) to find the eigenvalues and eigenvectors of $\mathbf{P}^T$.
   - The Python command `np.linalg.eig(T.transpose())` produces, for the eigenvalue 1, the eigenvector $\begin{bmatrix} -0.5974 & -0.7301 & -0.3319 \end{bmatrix}^T$.
   - Add up the values to get -1.6593.
   - Divide the eigenvector given by Python by this value to obtain the steady-state vector $\begin{bmatrix} 0.36 & 0.44 & 0.2 \end{bmatrix}$.
10. Remember that an eigenvalue has an *eigenspace*; thus, for instance, any scalar multiple of an eigenvector for the eigenvalue λ is also an eigenvector for λ.

iv. Example: Apply the steady-state idea to the gambling example
1. The theorem doesn't apply because the Markov chain isn't irreducible, but let's see what happens anyway.
2. We'll make the problem simpler by assuming that there are only four states: 0, 1, 2, 3, with $p = 0.5$.
3. This gives us the following one-step transition matrix:

$$\mathbf{P} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0.5 & 0 & 0.5 & 0 \\ 0 & 0.5 & 0 & 0.5 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

4. Via Python, we see that 1 is a double eigenvalue of $\mathbf{P}^T$, with corresponding eigenvectors $\begin{bmatrix} 1 & 0 & 0 & 0 \end{bmatrix}$ and $\begin{bmatrix} 0 & 0 & 0 & 1 \end{bmatrix}$.
5. This means that any linear combination of the vectors $\begin{bmatrix} 1 & 0 & 0 & 0 \end{bmatrix}$ and $\begin{bmatrix} 0 & 0 & 0 & 1 \end{bmatrix}$ is an eigenvector corresponding to $\lambda = 1$, and any convex combination of the vectors satisfies $\pi_0 + \pi_1 + \pi_2 + \pi_3 = 1$.
6. We don't get a unique steady-state vector.
7. This should make sense because the presence of the absorbing states means that, unlike the irreducible case, the long-term state of the system is dependent on the initial system state.

b. Expected average cost per unit time
i. We've already seen that if the Markov chain is either not irreducible or not ergodic, then the limiting probabilities may not exist.
ii. Let's take a closer look at the irreducible, finite-state (but not necessarily ergodic) case, though.
1. For example, suppose $\mathbf{P} = \begin{bmatrix} 0 & 0.2 & 0.8 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}$.
2. Then $\mathbf{P}^N = \mathbf{P}$ for all even powers of $\mathbf{P}$, and all odd powers of $\mathbf{P}$ are of the form $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0.2 & 0.8 \\ 0 & 0.2 & 0.8 \end{bmatrix}$.
3. There can be no limiting probabilities in this case, as the stochastic process oscillates back and forth between state 0 and one of state 1 and state 2.

iii. However, if a finite-state Markov chain is irreducible then the following exists: $\lim_{n\to\infty} \frac{1}{n} \sum_{k=1}^{n} p_{ij}^{(k)} = \pi_j$, where $\pi\mathbf{P} = \pi$ and $\sum_{j=0}^{M} \pi_j = 1$.
   1. Comments
      - This is the same vector π that gives the long-run probabilities of being in each system state.
      - Here, however, we see that the vector π also gives the long-run percent of time spent in each system state.
   2. Example: For our matrix $\mathbf{P}$, Python gives, after appropriate scaling, the eigenvector $\pi = \begin{bmatrix} 1/2 & 1/10 & 2/5 \end{bmatrix}$.

iv. Suppose it costs us \$5 every time we're in state 0, but we make \$8 and \$3, respectively, every time period we're in states 1 or 2. Do we make or lose money in the long run?
   1. Since the π vector gives the long-run percent of time spent in each system state, we can determine the expected (average) amount of money we make per time period in the long run by multiplying π by the vector $C = \begin{bmatrix} -5 & 8 & 3 \end{bmatrix}^T$.
   2. This is $-5/2 + 8/10 + 6/5 = -5/10 = -\$0.50$.
   3. Thus, with this system, we're losing 50 cents per time period in the long run.
   4. We would need to adjust our system if we're going to stay profitable.
   5. Formally, the long-run expected average cost per unit time for a Markov chain is given by $\lim_{n\to\infty} E\left[\frac{1}{n}\sum_{t=1}^{n} C(X_t)\right] = \sum_{j=0}^{M} \pi_j C(j)$.
   6. See text for how to deal with more complicated cost functions.

VI. First passage times
   a. Another aspect of a Markov chain that we sometimes care about is the set of first passage times.
   b. The *first passage time* from state i to state j is the number of time periods, starting in state i, it takes for the system to reach state j for the first time.
      i. For example, in the brewery example, we might be interested in the first time that someone who drinks brand A switches to brand B.
      ii. If $j = i$ then we have the *recurrence time* for state i, i.e., the number of time periods that it takes for the system to come back to i after first reaching it.
   c. The first passage times are random variables, and so we generally talk about *expected* first passage times.
   d. Let μ_{ij} denote the expected first passage time from state i to state j.
   e. Calculating expected first passage times
      i. If we condition on where the stochastic process is in the next time period we have $\mu_{ij} = \sum_{k=0,k\neq j}^{M} p_{ik}(1 + \mu_{kj}) + p_{ij}$.
      ii. Simplifying, this is $\mu_{ij} = \sum_{k=0,k\neq j}^{M} p_{ik}\mu_{kj} + \sum_{k=0}^{M} p_{ik} = 1 + \sum_{k=0,k\neq j}^{M} p_{ik}\mu_{kj}$.
   f. Expected first passage times into state A for the brewery example
      i. We have
         1. $\mu_{AA} = 1 + 0.25\mu_{BA} + 0.05\mu_{CA}$
         2. $\mu_{BA} = 1 + 0.75\mu_{BA} + 0.05\mu_{CA}$
         3. $\mu_{CA} = 1 + 0.1\mu_{BA} + 0.8\mu_{CA}$
      ii. Notice that only the equation with μ_{AA} on the left-hand side actually contains μ_{AA}.

iii. The solution to this system is $\mu_{AA} = 25/9$, $\mu_{BA} = 50/9$, $\mu_{CA} = 70/9$, or $\mu_{AA} \approx 2.78$, $\mu_{BA} \approx 5.56$, $\mu_{CA} \approx 7.78$.

iv. Thus, on average, it takes 2.78 months for someone to return to brand A after leaving it, 5.56 months for some to switch from brand B to brand A, and 7.78 months for someone to switch from brand C to brand A.

g. Expected recurrence times

i. Notice, in the previous example, that $\mu_{AA} = 25/9 = 1/\pi_A$.

ii. This is not a coincidence: In general, $\mu_{ii} = 1/\pi_i$.

iii. This should make sense, though.

1. If the probability of being in state i in the long run is π_i then the expected time to transition from i back to itself must be $1/\pi_i$.
2. Or think about it with respect to units: π_i is in occurrences per unit time, and μ_{ii} is in time per unit occurrences.

VII. Absorbing states

a. Yet another question that we're often interested in concerning Markov chains is the following: If there are absorbing states, what is the probability that we end up in a particular absorbing state?

- For example, in the gambling scenario, what is the probability that we go broke? What is the probability that we reach our goal?
- The answers generally depend on the initial system state.

b. The idea we use to determine the absorption probabilities is similar to that for finding the expected first passage times.

i. If f_{ik} is the probability that, starting in state i, we eventually end up in absorbing state k, then we have

$$f_{ik} = \sum_{j=0}^{M} p_{ij} f_{jk}.$$

ii. We also have:

1. $f_{kk} = 1$.
2. $f_{jk} = 0$ if state j is recurrent.

c. Let's find the probabilities of going broke in the gambling example with winning fortune \$3.

i. We have the following system of equations:

$$\begin{aligned} f_{00} &= 1 \\ f_{10} &= 0.5 f_{00} + 0.5 f_{20} \\ f_{20} &= 0.5 f_{10} + 0.5 f_{30} \\ f_{30} &= 0 \end{aligned}$$

ii. Solving for f_{10} and f_{20} we obtain $f_{10} = 0.5 + 0.5(0.5 f_{10}) \implies 0.75 f_{10} = 0.5 \implies f_{10} = 2/3$.

iii. Therefore we have $f_{00} = 1$, $f_{10} = 2/3$, $f_{20} = 1/3$, and $f_{30} = 0$.

iv. Thus, for example, the probability of going broke given that you start with \$2 is 1/3.

VIII. Continuous-time Markov chains

a. Idea

i. Thus far we've looked at stochastic processes indexed by discrete time steps.

ii. However, some situations aren't modeled very well using discrete time steps.

1. For example, queues really need to be modeled as continuous stochastic processes.
2. In this case the arrival and departure of customers happens continuously rather than in fixed time stages.

b. A continuous-time stochastic process is said to be *Markovian* if $P(X_{s+t} = j|X_s = i, X_r = k) = P(X_{s+t} = j|X_s = i)$ for all $t > 0$, $r \geq 0$, and $s > r$.
   - Once again, given the present, the past is irrelevant for predicting the future.

c. If these probabilities do not depend on s they are said to be *stationary*.
   - We will focus only on stationary transition probabilities.

d. Transition rates
   i. The probability p_{ij} doesn't make sense in the continuous case.
   ii. We can talk about $p_{ij}(t) = P(X_{s+t} = j|X_s = i)$; i.e., the probability that the stochastic process starts in state i and t time units later is in state j.
   iii. It's more useful to talk about the *transition rates* q_{ij}, though.
      - These are the number of transitions that the stochastic process makes from state i to state j per unit time, on average.
   iv. We can also talk about q_i, the transition rate out of state i.
      - We have $q_i = \sum_{j \neq i} q_{ij}$.

e. For example, consider a system of two machines and a maintenance person who works on them.
   i. Problem information
      1. Machine 1 breaks down μ_1 times per time unit on average.
      2. Machine 2 breaks down μ_2 times per time unit on average.
      3. The maintenance person can fix μ machines per time unit on average.
   ii. System states
      1. Both machines work
      2. Machine 1 has failed, machine 2 works, and maintenance person fixing machine 1
      3. Machine 2 has failed, machine 1 works, and maintenance person fixing machine 2
      4. Both machines have failed and maintenance person is fixing machine 1
      5. Both machines have failed and maintenance person is fixing machine 2
   iii. Graph of transitions and transition rates

f. Times between transitions
   i. We don't know exactly when the stochastic process will switch from one state to another, so the transition times are random variables.
      1. However, because of the stationarity and Markovian properties, the distribution of the time it takes to leave state i once it has been entered is the same regardless of when state i was first entered.
      2. Thus it makes sense to talk about the random variable T_i, the time that the stochastic process spends in state i before leaving it.

ii. Memorylessness
   1. Because of the Markovian property, though, we have $P(T_i > s + t | T_i > s) = P(T_i > t)$.
   2. This is called the *memoryless property*.
   3. There is only one continuous distribution that has the memoryless property; namely, the exponential distribution.
   4. Thus T_i is exponentially distributed.

iii. The exponential distribution takes a single parameter θ, which has the interpretation that $1/\theta$ is the mean.
   1. The mean amount of time that the process spends in state i should, intuitively, be the reciprocal of the transition rate q_i out of state i.
   2. Thus the mean of the distribution of T_i is $1/q_i$.
   3. Therefore, $T_i \sim exponential(q_i)$.

iv. Machines example
   1. The transition rates out of states 0 through 4 are, respectively, $\mu_1 + \mu_2$, $\mu + \mu_2$, $\mu + \mu_1$, μ, and μ.
   2. Thus the distribution of time that, for example, both machines are working is $exponential(\mu_1 + \mu_2)$.

g. Steady-state probabilities

i. For irreducible continuous-time Markov chains we can also talk about the long-run probabilities that the Markov chain finds itself in any particular state.

ii. Steady-state equations
   1. To be true steady-state probabilities they need to satisfy "rate out = rate in" for every state in the system.
   2. In equation form, this is $\pi_j q_j = \sum_{i \neq j} \pi_i q_{ij}$, for $j = 0, 1, \ldots, M$.
   3. And, of course, we must have $\sum_{i=0}^{M} \pi_i = 1$.

iii. Machines example
   1. The steady-state equations are
      - $\pi_0(\mu_1 + \mu_2) = \pi_1\mu + \pi_2\mu$
      - $\pi_1(\mu + \mu_2) = \pi_0\mu_1 + \pi_4\mu$
      - $\pi_2(\mu + \mu_1) = \pi_0\mu_2 + \pi_3\mu$
      - $\pi_3\mu = \pi_1\mu_2$
      - $\pi_4\mu = \pi_2\mu_1$
   2. This system is underdetermined; one equation is redundant.
   3. This missing equation is $\pi_0 + \pi_1 + \pi_2 + \pi_3 + \pi_4 = 1$.
   4. The solution to the system can be found by solving it in terms of a variable, say, π_4, and then scaling.
   5. For example, if $\mu = 3$, $\mu_1 = 1$, and $\mu_2 = 2$, then we have

$$
\begin{aligned}
3\pi_0 &= 3\pi_1 + 3\pi_2 \\
5\pi_1 &= \pi_0 + 3\pi_4 \\
4\pi_2 &= 2\pi_0 + 3\pi_3 \\
3\pi_3 &= 2\pi_1 \\
3\pi_4 &= \pi_2
\end{aligned}
$$

   6. Solving for this in terms of π_4 yields $\pi_0 = \frac{9}{2}\pi_4, \pi_1 = \frac{3}{2}\pi_4, \pi_2 = 3\pi_4, \pi_3 = \pi_4$.

7. Thus $\frac{9}{2}\pi_4 + \frac{3}{2}\pi_4 + 3\pi_4 + \pi_4 + \pi_4 = 1$, which yields
$$\pi_0 = \frac{9}{22},\ \pi_1 = \frac{3}{22},\ \pi_2 = \frac{6}{22},\ \pi_3 = \pi_4 = \frac{2}{22}.$$
8. Thus, on average (for instance) both machines are working $\frac{9}{22}$ of the time.
9. In general, the solution is

$$\pi_0 = \frac{\mu^2}{\mu^2 + \mu\mu_1 + \mu\mu_2 + 2\mu_1\mu_2},\ \pi_1 = \frac{\mu\mu_1}{\mu^2 + \mu\mu_1 + \mu\mu_2 + 2\mu_1\mu_2}$$
$$\pi_2 = \frac{\mu\mu_2}{\mu^2 + \mu\mu_1 + \mu\mu_2 + 2\mu_1\mu_2},\ \pi_3 = \pi_4 = \frac{\mu_1\mu_2}{\mu^2 + \mu\mu_1 + \mu\mu_2 + 2\mu_1\mu_2}$$

Homework: Handout problems #4-1, 4-5, 5-2, 5-7, 5-9, 6-2, 7-1, 8-2.